**Exp. No: 5**

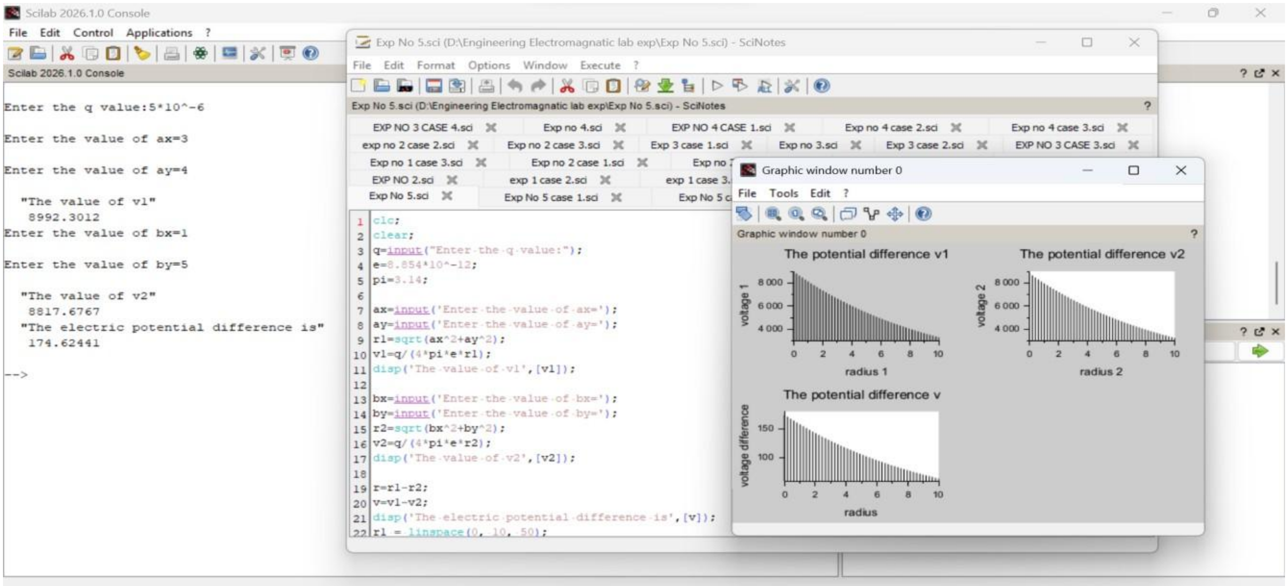
## Determination and plotting of Electric potential difference between any two points in free space medium separated by a distance R.

### Aim:

To write program for Electric potential difference between two points in free space medium using SCILAB.

### Software required:

• SCILAB version 6.0.1



EXP NO:- 05

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$$

A(3, 4)

$r = \sqrt{3^2 + 4^2}$

$= \sqrt{9 + 16}$

$= \sqrt{25}$

$= 5$

B(2, 5)

$r = \sqrt{1^2 + 5^2}$

$= \sqrt{1 + 25}$

$= \sqrt{26}$

$= 5.0990$

Point A electric Charge

$$V = \frac{8.99 \times 10^9 \times 5 \times 10^{-6}}{5}$$

$$= \frac{44950}{5}$$

$\boxed{V_1 = 8990\ V}$

$$V_2 = \frac{8.99 \times 10^9 \times 5 \times 10^{-6}}{5.0990}$$

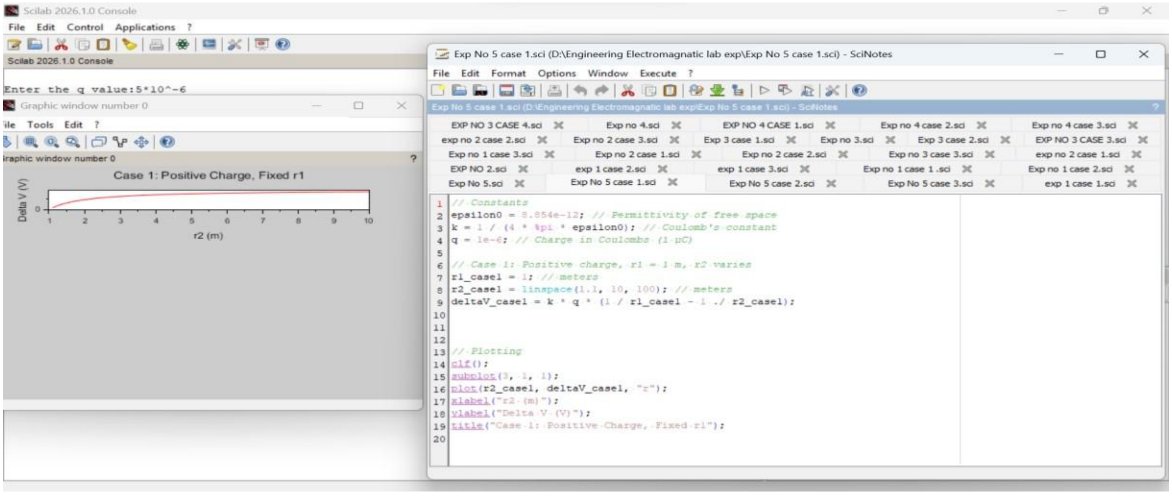
$$= \frac{44950}{5.0990}$$

$\boxed{V_2 = 8817.6\ V}$

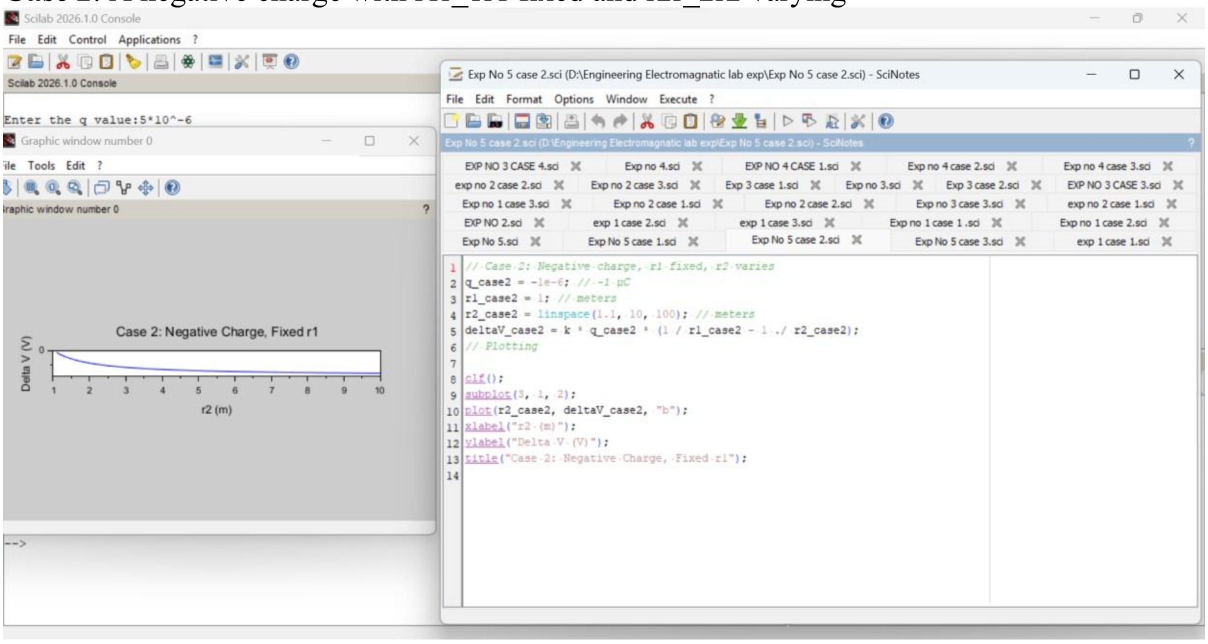
$\boxed{V_1 - V_2 = 174.60}$

O/P

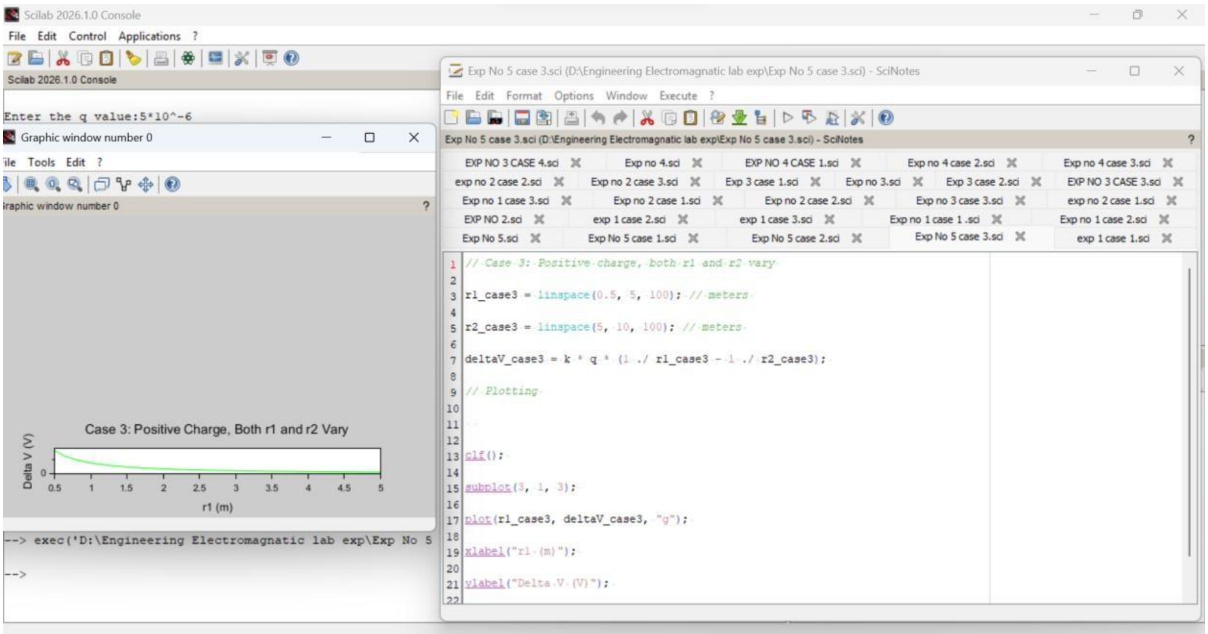
**Case 1**: A positive charge with fixed r1=1 m r2 varying.




**Case 2**: A negative charge with r1r_1r1 fixed and r2r_2r2 varying




**Case 3**: Both points r1r_1r1 and r2r_2r2 vary with a positive charge



**Result:**

Thus, the program was verified for Electric potential difference between two points in free space medium using SCILAB was successfully.